International Baccalaureate Diploma Program

Sports Psychology Extended Essay

To what extent does anxiety or lack of self-confidence negatively affect motor skills in athletes?

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Introduction

The athletic world has spread rapidly across the globe over the past century with the addition of many sports, such professional American football in the 1890s (Turpie, 2021) and basketball in 1891 (Springfield College, 2022). According to the Bureau of Labor Statistics, a part of the U.S. government which measures market activity (*Bureau of Labor Statistics | USAGov*, n.d.), around one-fifth of the United States played sports/exercised as an everyday habit in 2019 (Statista Research Department, 2023). Additionally, in 2020, more than half of U.S. children took part in sport-related activities (Black et al., 2022). Moreover, a year later, over fifty million viewers in the United States watched sports, whether highlights of games or live competitions, once or more every month (Statista Research Department, 2021). As shown in these three figures, athletic activities are at least rather prevalent in modern society.

Athletic activities demand an abundance of physical exertion to perform optimally, as most sports require many explosive movements (i.e. sprinting and jumping). However, there is a side to sports that athletes tend to neglect: the mental side. According to Chris Mance II, a sport psychology coach (Mance II, 2018), "...playing sports is 100% mental," (Mance II, 2021). Because of this, athletes need to also be aware of the mental side of their game to perform at their best. This can be difficult due to the amount of mental and physical pressure athletes are put under during performances, leading to mental health conditions, such as anxiety or depression, or even a lack of self-confidence. Weber, Samantha R. et al published in January 2023 found that 22.3 percent of National Collegiate Athletic Association (NCAA) Division I and II players were susceptible to depression, 12.5 percent to anxiety, and eight percent to low self-esteem (p. 1211). In the fall of 2021, a survey found that around a quarter (24%) of male and over a third (36%) of female athletes "felt so depressed that it was difficult to function," (NCAA). Among professional

athletes in sports such as cricket and rugby, 35 percent suffer from a mental health crisis that can be expressed through persistent depressive symptoms (Reardon et al., 2019, pp. 667-699).

With the prevalence of mental health struggles in many different student, collegiate, and professional athletes, it is important to discuss the effects that anxiety or a lack of self-confidence may have on an athlete's motor skills, which is why in this paper, I will attempt to answer the question: "To what extent does anxiety or a lack of self-confidence negatively affect motor skills in athletes?".

Since an athlete's motor skills have a direct effect on their ability to perform, I will often use an athlete's sports performance to measure the effect that anxiety or a lack of self-confidence has on their motor skills.

Motor skills refer to the nervous system's ability to facilitate body movement, and are divided into two categories: gross and fine motor skills (Sutapa et al., 2021). Gross motor skills are made up of "...locomotor, object control, and balance skills," (Sutapa et al., 2021; Bardid et al., 2016), and utilize large muscles (Kokštejn et al., 2017). Fine motor skills include coordinated motion between multiple organs (hand-eye coordination, for example), and also finger movement (Carlson et al., 2013). Motor performance itself combines motor skills and "...fitness-related components," (Vallence et al., 2019, p. 1-8), such as coordination, balance, and speed (Biddle et al., 2015, p. 10). According to Montana State University Athletics, it is widely agreed upon that mental health challenges can impact an athlete's ability to perform optimally (DeStefano, 2021), making it important to study to help athletes improve their motor skills. Having previously experienced anxiety/nervousness and a lack of self-confidence as an athlete myself, I hypothesize that all forms of anxiety, as well as a lack of self-confidence, can significantly impair an athlete's motor skills, and therefore, performance.

What is anxiety and self-confidence?

Anxiety can be defined as a long-lasting mental state of uneasiness caused by uncertain, and thus, potentially threatening future events (Knight et al., 2019). This particular mental health condition can be broken down into four different categories: state and trait anxiety, and cognitive and somatic anxiety.

State anxiety can be defined as a temporary reaction to adversity, meanwhile, trait anxiety consists of the constant tendency to respond to potentially distressing situations with concerns or worries (Saviola et al., 2020) and is thought to be a personality feature (Spielberger, 1970). Symptoms of state anxiety include a rapid heartbeat, poor concentration, intense worry, and muscle tension. Contrarily, those with trait anxiety may feel more anxious in everyday situations, like not receiving a text reply from a friend for a few hours and thinking they did something wrong and are being ignored (Khiron Clinics, 2022).

Cognitive anxiety can be defined essentially as “mental anxiety” composed of specific mental processes that occur when one is anxious, such as concern or worry (Gelenberg, 2000, pp. 49-54). Somatic anxiety, though, is instead “physical anxiety”, which consists of the way in which anxiety is exhibited in one’s body, and “is often accompanied by specific symptoms associated with a state of autonomic arousal, such as sweating, dizziness, and shortness of breath,” (Martens et al., 1990).

Self-confidence, while not a mental health condition like anxiety, is also a mental state that can facilitate an athlete’s motor skills during competition. It is created through one’s well-being, acceptance of mind and body, and belief in their ability (Budin, 2017). It stands to

reason, then, that self-confidence and lack thereof may affect one's ability to perform in athletic situations due to it in large part dealing with the ability to believe in oneself.

Evaluation of anxiety and self-confidence in athletes

Since anxiety and self-confidence are psychological constructs in human beings, preventing them from being studied directly, there needs to be a clear method to evaluate them in athletes. Many studies conducted on this topic have two common methods used to identify the level of both anxiety and self-confidence in athletes.

One method that will be used to evaluate both anxiety and self-confidence in athletes throughout this paper is the Competitive State Anxiety Inventory-2 (CSAI-2), the most commonly used scale in sports psychology for assessing anxiety levels (Ağaoğlu, 2016). This scale was developed by Rainer Martens, a former Professor of Kinesiology at the University of Illinois and founder of Human Kinetics (Martens, Rainer (1977-2012) | University of Illinois Archives), and colleagues in 1990. It is made up of 27 items (statements with answers from 1 to 4 on a Likert scale), which evaluate cognitive anxiety, somatic anxiety, and self-confidence intensity levels (Dingley, 2021). A Likert scale, invented by American social scientist Rensis Likert, typically uses a 5 or 7-point answer range to get an understanding of the feelings of those who answer the questions (Khosravi, 2023). The final score of the inventory is a number between 9 and 36. The closer that number is to 9, the lower the anxiety and the higher the self-confidence, whereas scores closer to 36 indicate the opposite.

Another method that will be used for evaluation is the Spielberger State-Trait Anxiety Inventory (STAI), which measures state and trait anxiety. Form Y, the most popular version of

the inventory, contains 40 items for evaluating state and trait anxiety: 20 items assess each (Spielberger et al., 1983). These items use 4-point Likert scales, with the total scores ranging from 20 to 80 (Spielberger et al., 1970), which can then be classified as “no or low anxiety” (scores of 20-37), “moderate anxiety” (scores of 38-44), and “high anxiety” (scores of 45-80) (Kayikcioglu et al. 2017).

While both of these methodologies do give decent insight into how anxious or confident an individual is feeling, anxiety and self-confidence are ultimately psychologically created constructs that cannot be quantified.

Oftentimes, while evaluating anxiety and self-confidence, researchers will look at both the intensity and direction of these mental states. Intensity touches on the level of anxiety or self-confidence that athletes actually possess, while direction refers to whether an athlete interprets that level of anxiety or self-confidence “...as either debilitative or facilitative…in the context of sports motor performance,” (Jones et al., 1994, pp. 657-663). Debilitative anxiety sees stresses as dangers and hindrances, often leading to worse performance, whereas facilitative anxiety sees threats as opportunities, and therefore, boosts performance (Nanda, 2023).

Effects of anxiety and lack of self-confidence on athletes’ motor performance

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Horikawa, M., & Yagi, A. (2012)

Horikawa, M., & Yagi, A. (2012) aimed to test how the level of trait anxiety influences state anxiety, and thus, penalty shoot-out performance in high pressure situations. The researchers incorporated the STAI to select groups of athletes with high and low anxiety out of

two groups of eight male soccer players from Kwansei Gakuin University in Japan, who all performed 20 penalty shots without a goalkeeper. The researchers also created high and controlled pressure conditions through the use of manipulated instructions. Each condition had 10 trials in a within-subject, or repeated measures, design, where all participants took part in every condition and the results were then compared (Bhandari, 2021). The dependent variable was the number of successful goals each player had, as well as the state anxiety scores in each condition (Horikawa & Yagi, 2012).

Findings

The mean trait and state anxiety scores for the low anxiety groups (LAGs) were 31.63 and 37.25 respectively in controlled conditions and 37.75 for the state anxiety score under pressure versus 54.00 and 41.25 for the high anxiety groups (HAGs) in controlled conditions and 44.88 for the state anxiety score under pressure. The number of successful goals for the LAGs was an average of 6.25 in comparison to 4.13 in the HAGs. The pressure conditions also showed a significant difference in the number of successful goals with 5.50 compared to 3.75 in the LAGs versus HAGs. As shown, the state anxiety scores were higher and the number of successful goals was much lower in the high than in the low-trait anxiety groups in both control and pressure conditions. This suggests that players with higher trait anxiety scores were more likely to then experience increased state anxiety in a high-pressure situation, leading to an interference with goal performance. A repeated measures analysis of variance (ANOVA), a method used to separate variance data into different components for further testing (Kenton, 2021), and a Bonferroni test revealed that the main effect, which refers to the variable of primary interest to the researchers (Montgomery, 1976), of the instructional condition was significant (p

< .01), and the number of successful goals was significantly lower ($p < .01$) in the pressure than in the control conditions for both low and high-trait anxiety players (Horikawa & Yagi, 2012).

Evaluation

This study is valuable because it was ethically approved by the Department of Psychological Sciences of Kwansei Gakuin University, while also following the standards of the American Psychological Association. Also, it utilizes many different studies directly related to this topic for reference, such as the highly influential Hardy and Parfitt (1991), which sought to test Fazey and Hardy's (1988) catastrophe model. However, this study only contains a total of 16 participants who are all male university soccer players, limiting its generalizability. Additionally, this study only measures goal performance during a penalty shoot-out without a goalkeeper rather than in live soccer matches.

Analysis

It can be deduced from this study that trait anxiety has a larger influence than state anxiety because the findings suggested that players with higher levels of trait anxiety then had higher levels of state anxiety. The number of successful goals in the LAC (6.25) was 51.3% higher than that of the HAC condition (4.13), meaning that a clear difference in performance could be seen even before pressure was added. Additionally, regardless of whether an athlete had low or high-trait anxiety, they were significantly affected when it came to pressure conditions ($p < .01$). It can then be concluded that trait anxiety has a significant negative impact on the motor skills of a soccer athlete both with and without pressure.

Chamberlain, Sean T., and Bruce D. Hale (2007)

Chamberlain, Sean T., and Bruce D. Hale (2007) aimed to correlate anxiety to accuracy on a golf-putting task (target was 5 meters away). Anxiety was measured by the CSAI-2(D), a modified version of the CSAI-2, in 12 seasoned male golfers. Unlike the original CSAI-2, this inventory also includes a direction scale for respondents to rate the extent to which anxiety was either helpful or harmful to performance (Jones & Swain, 1992, pp. 467-472). The scores for anxiety and putting performance in three anxiety-controlled conditions were used to evaluate the predictions of the Multidimensional Anxiety Theory (MAT; Martens et al., 1990), which predicts that "cognitive and somatic anxiety will differentially and independently relate to performance," (Krane, 1990; Chamberlain & Hale, 2007).

Findings

The intensity of cognitive anxiety exhibited a negative linear relationship with putting scores ($p < .01$), the intensity of somatic anxiety showed a curvilinear relationship ($p < .05$), and the intensity of self-confidence showed a positive linear relationship ($p < .05$). However, direction may be more important to analyze as it elucidated 42.4% of performance variance in comparison to 22.9% by the variables' intensity. Cognitive anxiety direction accounted for 18.9% ($p < .01$) of the variance linearly while intensity explained 22.4% ($p < .01$), somatic anxiety direction had no interpretable trend as either a linear equation or quadratic function (for both, $p > .05$) while intensity totaled 23.5% ($p < .05$), and finally, self-confidence direction accounted for 30% ($p < .01$) of the variance within a linear equation while intensity demonstrated 16.8% ($p < .05$) (Chamberlain & Hale, 2007).

Evaluation

This study is valuable because it uses putting, which only requires good motor performance to properly execute, allowing for the effects of anxiety to be measured accurately. Additionally, the researchers determined anxiety using the CSAI-2(D), which is based on the previously validated CSAI-2 (Lever et al., 2020, pp. 1-9). However, this study contains only 12 experienced male golfers, excluding many potential players, limiting its generalizability. The study also provides little information on the golf-putting results themselves but rather focuses on what tool within the MAT will better predict putting performance.

Analysis

The significantly negative linear relationship that cognitive anxiety intensity demonstrated means that, unwaveringly, more cognitive anxiety led to worse motor performance. Somatic anxiety intensity's curvilinear relationship means that more somatic anxiety improved motor performance up to a certain point before a continued increase led to poor putting performance. Self-confidence intensity showed a significant positive linear relationship with performance, meaning more confidence led to better motor performance. Additionally, the direction of cognitive anxiety and self-confidence also having significant relationships with performance means that the interpretation of each also has a large impact. With each of these metrics having a significant relationship with an athlete's performance (other than somatic anxiety direction), it can be concluded, then, that cognitive/somatic anxiety and a lack of self-confidence, whether through intensity, direction, or both, can have a significantly negative impact on an athlete's motor performance in golf.

Parfitt, G., & Pates, J. (1999)

Parfitt, G., & Pates, J. (1999) aimed to test the influence of competitive anxiety and self-confidence state responses on performance. The researchers selected 12 basketball players from the University of Wales in England who self-reported their anxiety and self-confidence as a single response through the CSAI-2, and on a separate scale of 0 (low) to 27 (high) multiple times before playing. The players' performances were video-recorded, and the researchers also measured largely anaerobic (i.e. jump height on rebounds) and working memory aspects of performance (i.e. completed passes and assists). They conducted regression analyses on data gathered from seven matches to see which variables were impactful (Gallo, 2015), and hierarchical regression analyses to account for all other variables (Kim, 2016; Parfitt & Pates, 1999).

Findings

Somatic anxiety predicted aspects of performance that involved anaerobic demands positively, while cognitive anxiety was not the main predictor of working memory aspects of performance, but rather self-confidence. Each state response had a significant prediction of the variance across all performance scores ($14\% \leq X \leq 65\%$, where 'X' is the percentage predicted by each state response). Independently, somatic anxiety predicted the most amount of variance on the rebound jump with 38%, self-confidence predicted the most on successful passes with 51%, and also on assists with 65%. After the hierarchical regression analyses were conducted, however, they found that somatic anxiety marginally predicted 6% of the variance in height jumped on rebounds (not significant as $p < .08$), self-confidence predicted 34% in successful passes ($p < .01$), as well as 57% in assists ($p < .01$) (Parfitt & Pates, 1999).

Evaluation

This study is valuable because using quantitative data through multiple regression analyses allowed researchers to determine a cause-and-effect relationship between anxiety, self-confidence, and subsequent performance in the players. Also, this study contains researcher triangulation since it uses two researchers (Parfitt and Pates), thus reducing the chance for bias and allowing for more interpretation of the data. However, it only contains 12 participants, who are all from the University of Wales, limiting this study's generalizability. Additionally, Parfitt and Pates used three metrics for determining performance (height jumped, completed passes, and assists) while many others may have been more appropriate to use, such as shooting, which requires extreme accuracy.

Analysis

In Parfitt, G., & Pates, J. (1999), passing, assisting, and rebounding all require many motor skills. However, their findings do not suggest that somatic anxiety has a dramatic effect on motor performance after factoring in other variables, as predicting only about 6% of the variance of the total jump height ($p < .08$) after a hierarchical regression analysis. Self-confidence predicted a significant 34% of the variance in successful passes ($p < .01$), and 57% of the variance in assists ($p < .01$) also after the hierarchical regression analysis, demonstrating how impactful self-confidence can be on an athlete's motor performance regardless of other responses. It can then be concluded from this study that both somatic and cognitive anxiety have an insignificant impact on performance including other state responses, while self-confidence is significant in determining ability to complete athletic tasks involving working memory.

Hardy, L., Parfitt, G., & Pates, J. (1994)

Hardy, L., Parfitt, G., & Pates, J. (1994) aimed to test Fazey and Hardy's (1988) catastrophe model of anxiety and performance, which proposes cognitive anxiety as the splitting factor between whether the effect of physiological arousal (the normal factor) is small, catastrophic, or somewhere in between, based on the perception of the arousal symptoms by athletes/performers (Fazey & Lew Hardy, 1988; Hardy et al., 1994). The study attempted to test this model using eight seasoned crown green bowlers who bowled in conditions of high (HCA) and low cognitive anxiety (LCA). Crown green bowling is a form of bowling played on a convex, uneven grass surface (Williams, 2022a) where a smaller bowl, known as the jack, is "delivered" (bowled) before the players' bowling balls, and the objective is to get their bowl as close to that jack as possible (Williams, 2022b). In each condition, physiological arousal (heart rate) was increased and decreased through physical work. A repeated-measures three-factor ANOVA test revealed that cognitive anxiety, heart rate, and the direction that the heart rate changed upon performance had a three-way interaction (Hardy et al., 1994).

Findings

Through correlated, or paired, t-tests, a method used "...to determine whether the mean difference between two sets of observations is zero," (*Paired T-Test*, n.d.), the researchers found that in the HCA condition, the athletes' best performances were significantly better ($p < .005$) and their worst performances were significantly worse than in the LCA condition ($p < .01$). In the HCA, heart rate increasing (HRI) group (HCAHRI), the average error score (the average distance each bowl was from the jack measured in meters ('x') then transformed using $f(x)=1/x$) was 0.736 ± 0.334 with an outlier of 1.33 ± 0.75. In the HCA, heart rate decreasing (HRD)

group (HCAHRD), the average error score was 0.532 ± 0.122. In the LCAHRI group, the average error score was 0.544 ± 0.108. Lastly, in the LCAHRD group, the average error score was 0.526 ± 0.110. However, these did not explicitly support the catastrophe model (Hardy et al., 1994).

Evaluation

Hardy, L., Parfitt, G., & Pates, J. (1994) is valuable because it contains three different researchers, reducing the chance for bias and allowing for more interpretation of the data, thus making the study more credible. Additionally, this study refers to lots of information from past research carried out on this topic, such as Fazey and Hardy (1988), Burton (1988), and Martens et al (1990), further increasing credibility. However, this study uses only one sport, which is not widely played around the world, limiting its external validity. Also, this study only uses eight participants, who are all experienced in their sport, dramatically limiting its generalizability.

Analysis

The HCAHRI group was the only group to display a significant increase in the number of errors with an average error score of 0.736 ± 0.334. This is much higher than the HCAHRD and LCAHRI groups where the average error score was 0.532 ± 0.122 and 0.544 ± 0.108 respectively. This suggests that the HCA groups were more inconsistent with their performances based on whether their heart rates were increasing or decreasing while the LCA groups had similar amounts of errors despite their level of physiological arousal (0.544 ± 0.108 high versus 0.526 ± 0.110 low). Based on this information, the amount of physiological arousal an anxious

athlete experiences can dramatically impair their consistency with their motor performance, while a non-anxious athlete may not experience a difference.

Discussion

Horikawa, M., & Yagi, A. (2012) discovered athletes with more trait anxiety often had more state anxiety, which significantly decreased their performance in both control and pressure conditions. Chamberlain, Sean T., and Bruce D. Hale (2007) found increased cognitive anxiety had a negative correlation with performance, while increased levels of perceived self-confidence correlated with performance positively. Parfitt, G., & Pates, J. (1999) found that self-confidence best predicts performance, meaning the most prominent factor, in tasks involving working memory (tasks requiring quick decision-making, such as assisting in basketball) compared to somatic and cognitive anxiety. Hardy, L., Parfitt, G., & Pates, J. (1994) found that high cognitive anxiety coupled with high amounts of physiological arousal led to more errors in an athlete's motor performance than if their arousal was low, or if they had low cognitive anxiety.

Common limitations of these studies included a lack of live performance competition and minimal potential for applying these results to outside groups since many participants were selected from an atypical location, or very few were selected. However, common values included a good number of researchers in each study, as well as many references to other studies done in the past on similar topics, and the use of previously established psychological tools, such as the STAI and the CSAI-2, for determining levels of anxiety and self-confidence.

Parfitt, G., & Pates, J. (1999) and Hardy, L., Parfitt, G., & Pates, J. (1994) both tested the effects of anxiety and/or self-confidence in real competition situations, but Horikawa, M., &

Yagi, A. (2012) and Chamberlain, Sean T., and Bruce D. Hale (2007) both only tested the effects of anxiety and/or self-confidence in practice scenarios. This may then make it unclear as to how an athlete's motor performance may be affected in live competitions. However, through pressure manipulation in Horikawa, M., & Yagi, A. (2012), it can be assumed that in live performance competitions (i.e. a live basketball game), athletes will experience impaired motor performance when anxious. Additionally, since self-confidence was significant in predicting performance during the basketball games in Parfitt, G., & Pates, J. (1999), it can then be assumed that the same will happen in live performance situations, where higher self-confidence leads to higher performance.

I had previously hypothesized that all forms of anxiety and a lack of self-confidence could significantly impair an athlete's motor skills, and therefore, performance, and after analyzing the literature, it appears that all forms of anxiety and a lack of self-confidence can significantly impact an athlete's motor performance in most circumstances. This means that my hypothesis can be accepted.

Overall, it can be deduced that, regardless of the specific type, higher anxiety negatively correlates with performance, and higher self-confidence improves motor performance in athletes.

Conclusion

As expressed throughout this paper, anxiety and a lack of self-confidence can both play a role in reducing the ability of an athlete to execute motor skills accurately during competition. As mentioned earlier, sports is mostly a mental game, making it necessary for athletes to train their minds as well. To conclude, having cognitive or somatic anxiety, or a lack of self-confidence

leads to worse athletic performance to a significant extent, while trait and state anxiety also may have a significant negative effect but there is limited research. Thus, this conclusion then calls for further research to be done on state and trait anxiety's impact on athletic motor performance. Other research that can be conducted on this topic includes, if ethically possible, the effects of depression on an athlete's motor performance. Providing such research would then make the importance of maintaining good mental health more clear to athletes to thus improve their consistency in competition.

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